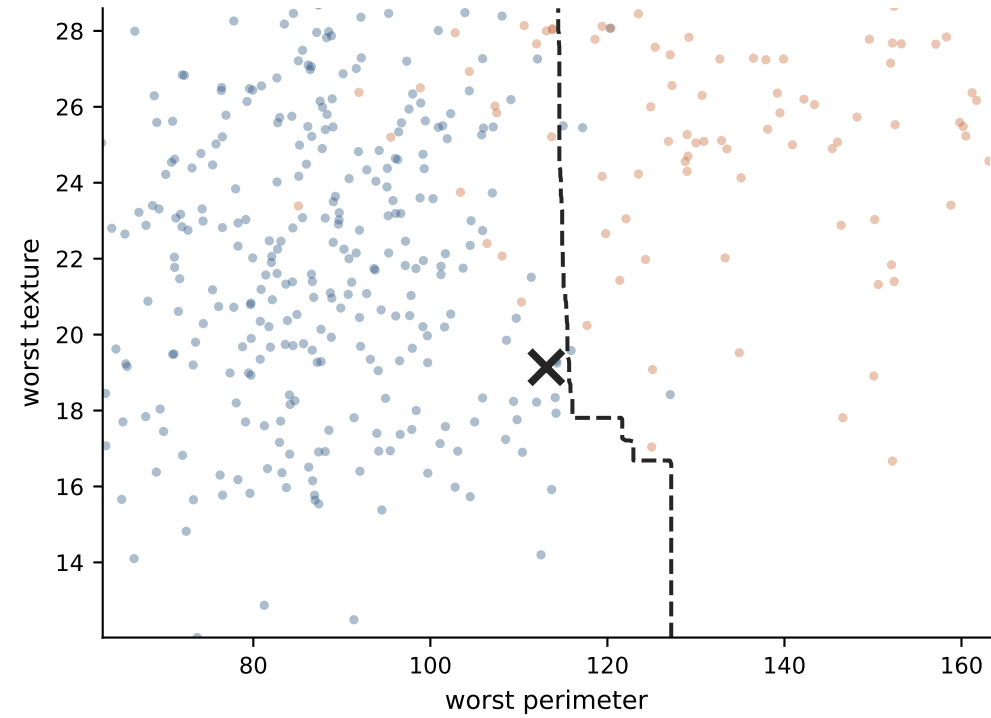


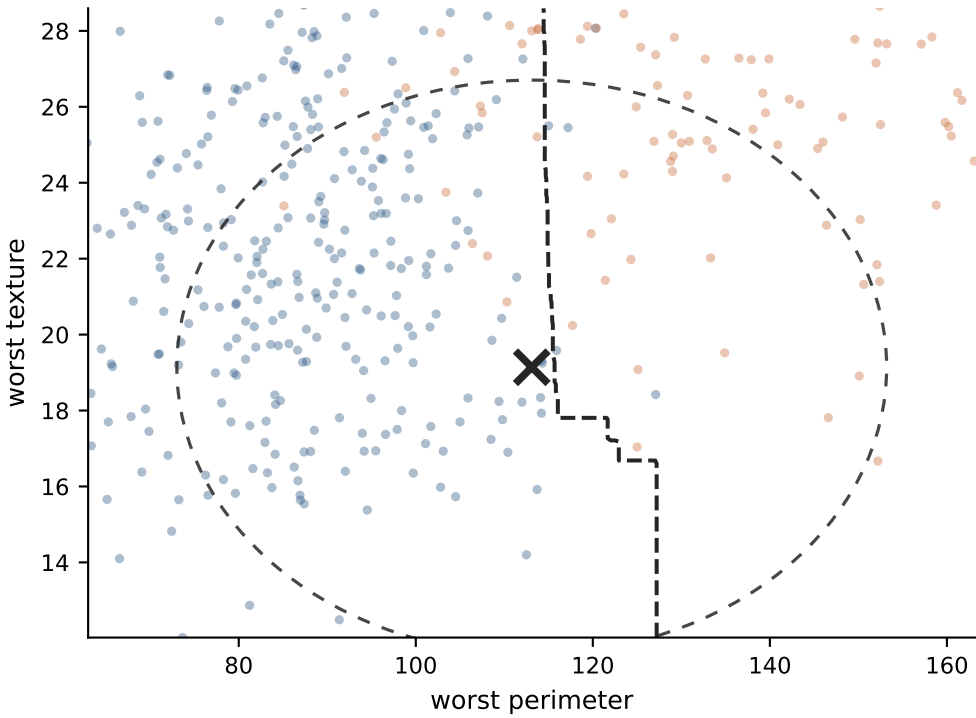
# LIME

Local Interpretable Model-agnostic Explanation — patient #67,  $P(\text{benign})=0.581$ ,  $R^2=0.36$

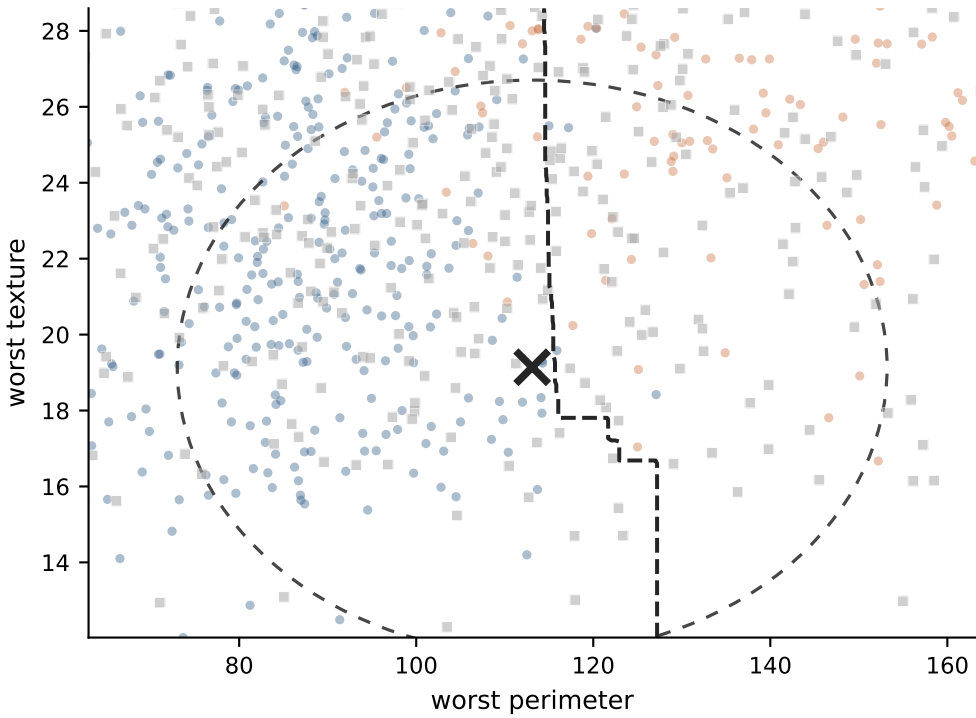
**A · the black-box model**



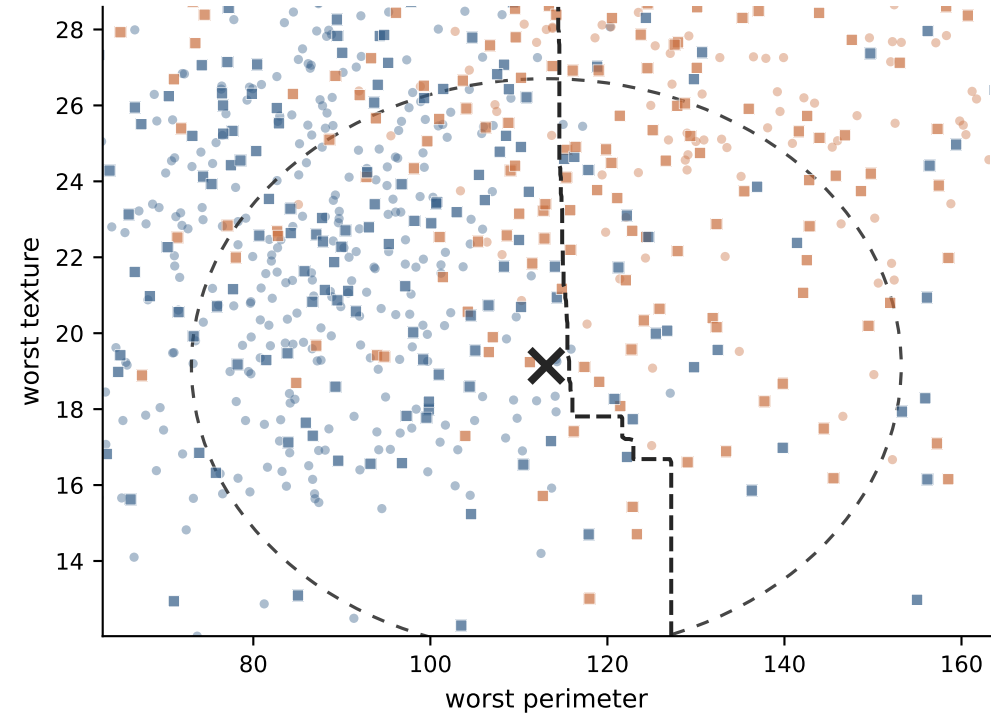
**B · the local neighborhood**



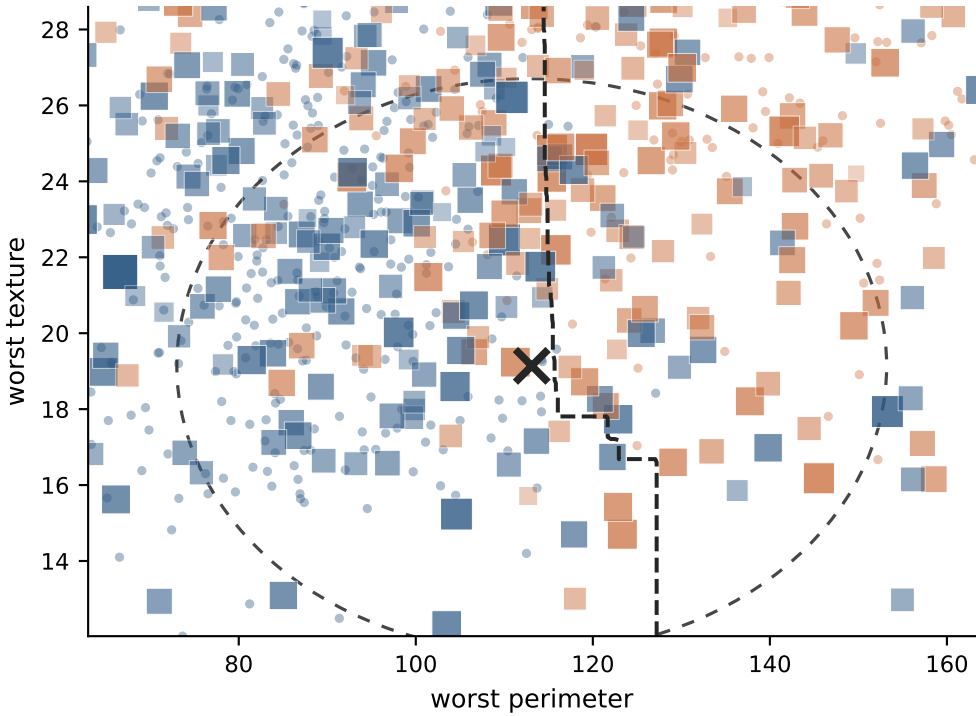
**C · perturbation**



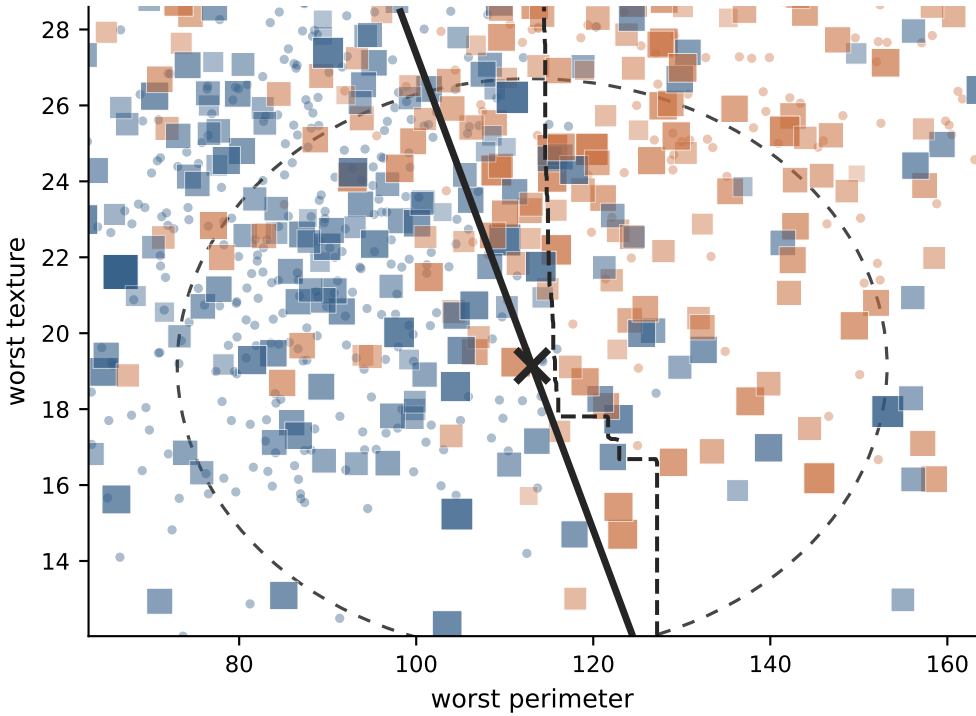
**D · predictions on the neighbors**



**E · proximity weighting**



**F · the local fit**



● real patient · benign  
● real patient · malignant

■ neighbor · predicted benign  
■ neighbor · predicted malignant

--- model boundary ( $P=0.5$ )  
---  $\pm 1.2\sigma$  reference

✕ patient #67  
— local fit (through the patient)